

Hyperparameter : Search/optimization



KNN
↳ (k) Best

↳ $\lambda \rightarrow \lambda = 0 \rightarrow \text{overfitting}$

$\lambda \rightarrow \infty \rightarrow \text{underfitting}$

Q1
→ (i) GridSearchCV
→ (ii) RandomSearchCV

(case i)

ML($\lambda = n$)

for i in λ
ML($\lambda = i$)

Note → Acc

↓
→ $\lambda =$ final

$\lambda = [0.001, 0.01, 0.1, 1, 10]$

(Case ii)

(5) m^k time
2 min
12 min
6-2

ML ($\lambda_1 = n_1 \rightarrow \lambda_2 = n_2$)
 $\lambda_1 = ?$
 $\lambda_2 = ?$

eg

$\lambda_1 = [0.001, 0.01, 0.1, 1, 10]$
 $\lambda_2 = [0.001, 0.01, 0.1, 1, 10]$

~~interview~~

interview

for i in λ_1
for j in λ_2
ML ($\lambda_1 = i, \lambda_2 = j$)
all

Grid search

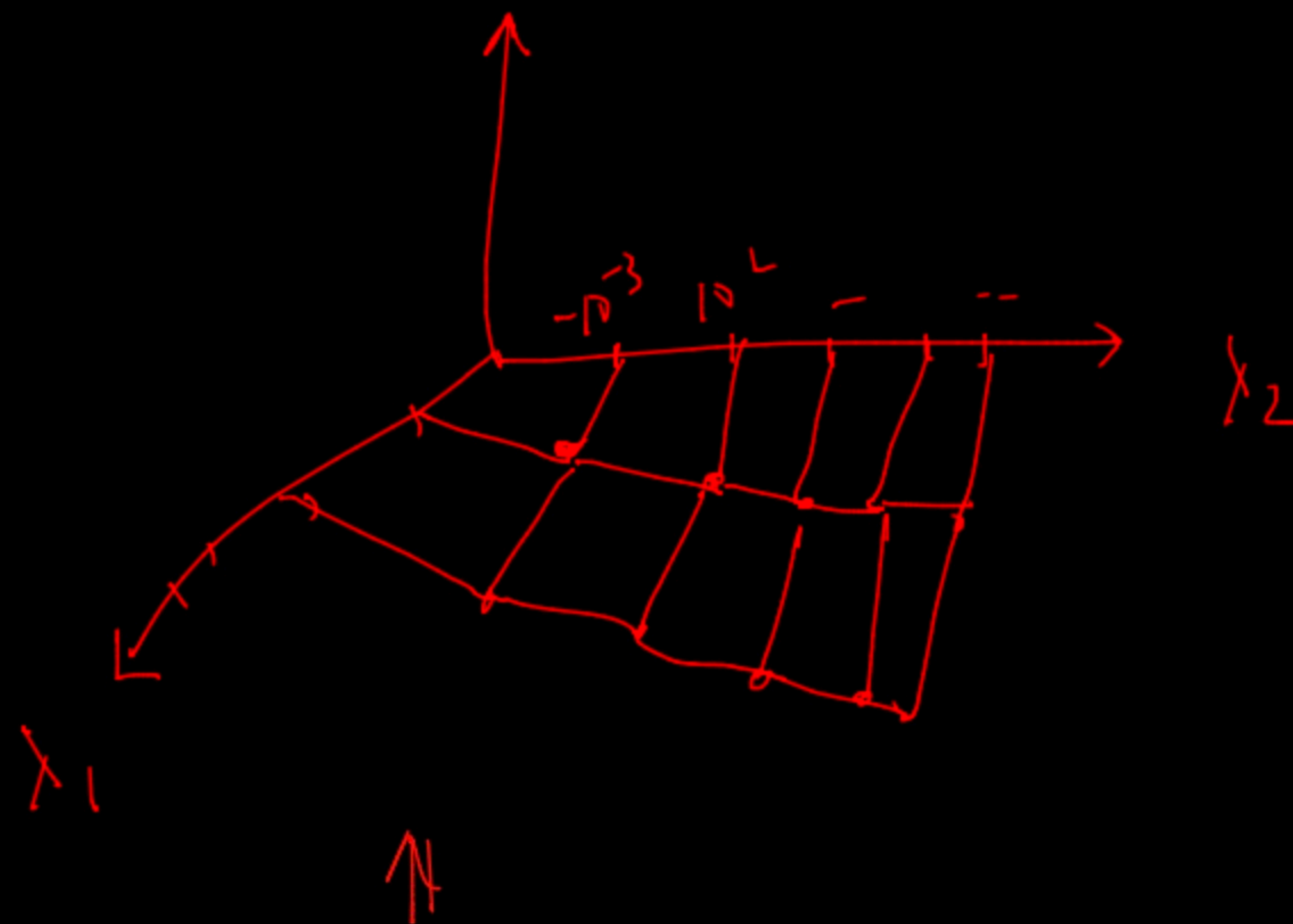
$\lambda_1 \rightarrow 1$ parameter
 $\rightarrow m$ times

$\lambda_1 \lambda_2 \dots m_1 \times m_2 \dots$ time

$\lambda_1 \lambda_2 \lambda_3 = m_1 m_2 m_3$ time

k parameter $\Rightarrow m^k$ time

parameter $\rightarrow$ # of time
model need to be trained
increasing exponentially



(i) RandomSearchCV

$$X = [10^{-4} \text{ --- } 10^4]$$

↑

↑ Range

→ pick a random value

(BS) ↓

(LR)

(ii) Bayesian optimization

↑

Hyperopt

↑

sklearn

Linear Regression

Actual Regression technique

eg,

	X			Y
	discount	B141	weekend	Sales
	0.12	-	-	-
	1	1	1	1
	1	1	1	1
	1	1	1	1

(x, y) $\rightarrow$ Real

	f_1	f_2
$\checkmark$	0	0
$\checkmark$	0	0
$\checkmark$	0	0

$$W = [\text{0.1} \quad 0.1] \text{ --- MLE } \rightarrow ?$$

$$y = w^T \cdot x_i + w_0$$

Step

(4)

$$LR(\underline{x}, y) \rightarrow y$$

$$\underline{Sales} = \underline{w_1} * \underline{discount} + \underline{w_2} B141 + w_3 * w + w_0$$

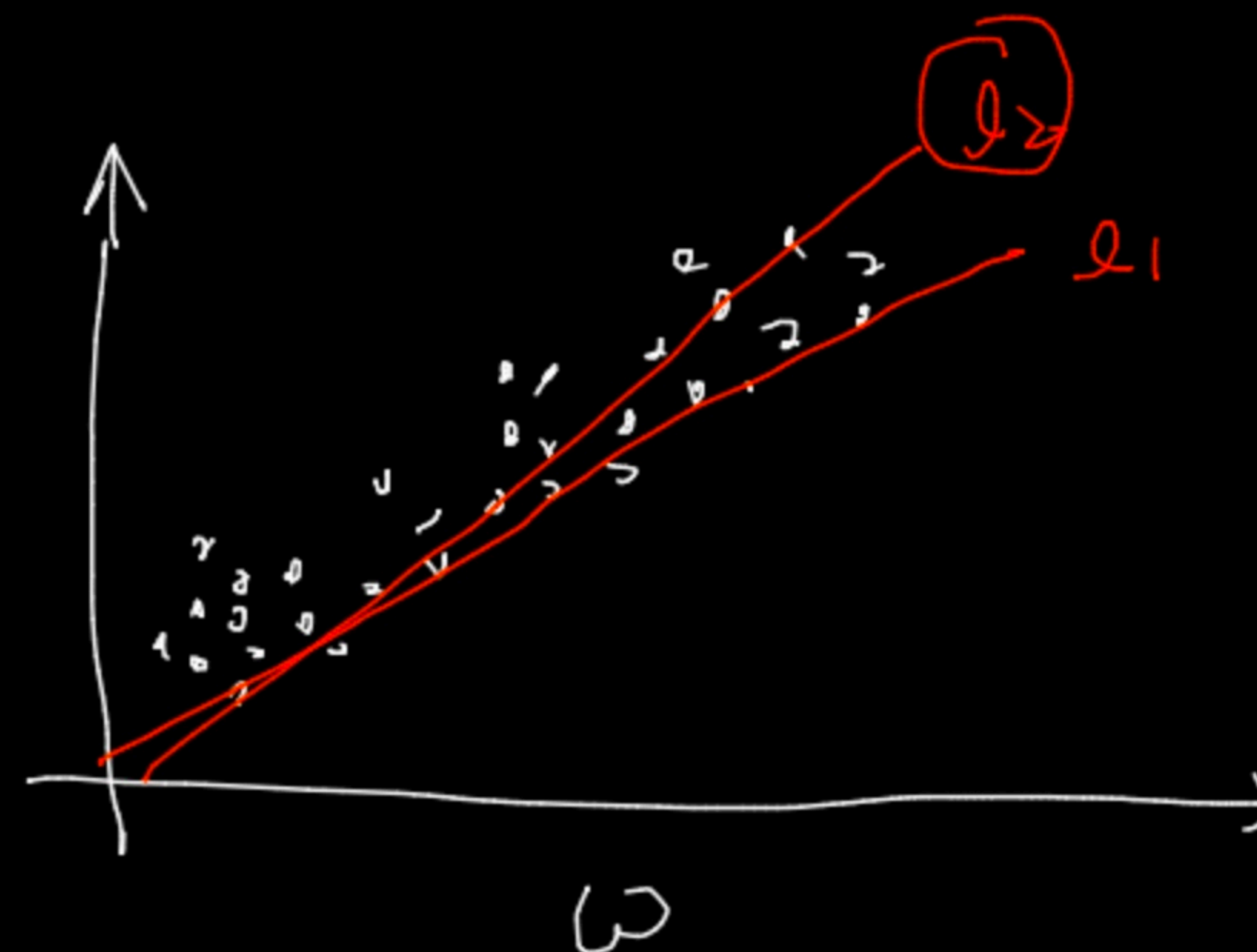
height $\rightarrow$ weight
 $\uparrow$

data

Lin Regression

w	h
$\textcircled{1}$	$\textcircled{0}$
...	...

plot $\rightarrow$



Lin Regression
 $\Downarrow$

Find out
best fit

$\Downarrow$
Training

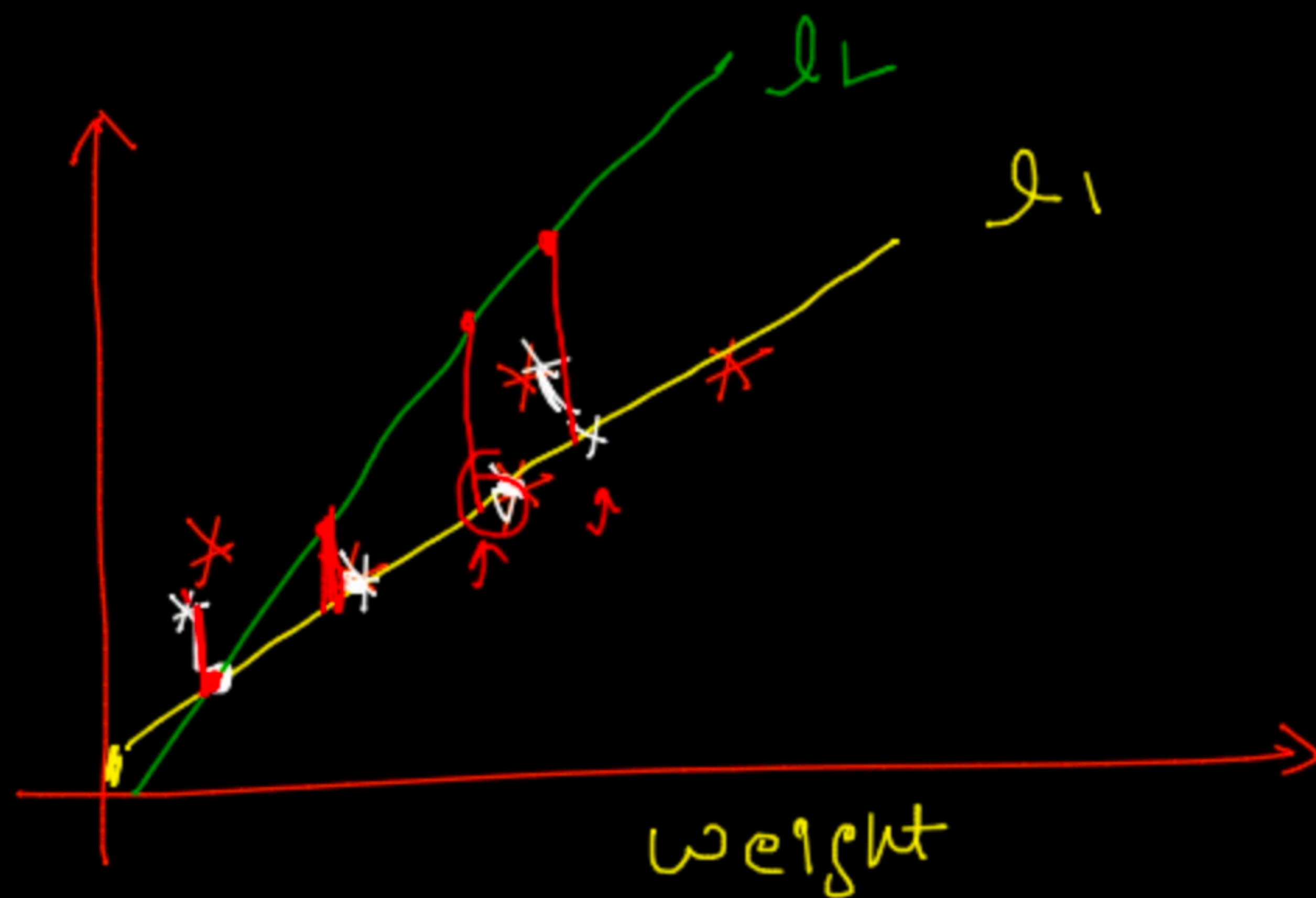
line
 $\rightarrow$ which will give me min error

$\mathcal{L}_1$

$\mathcal{L}_1$

w_1

$$h = 0.5 + w_1 x + w_0$$



(MSE)
↑

$\mathcal{L}_1 \Rightarrow$

$$h = 0.1 \times \text{weight} + 0$$

testing

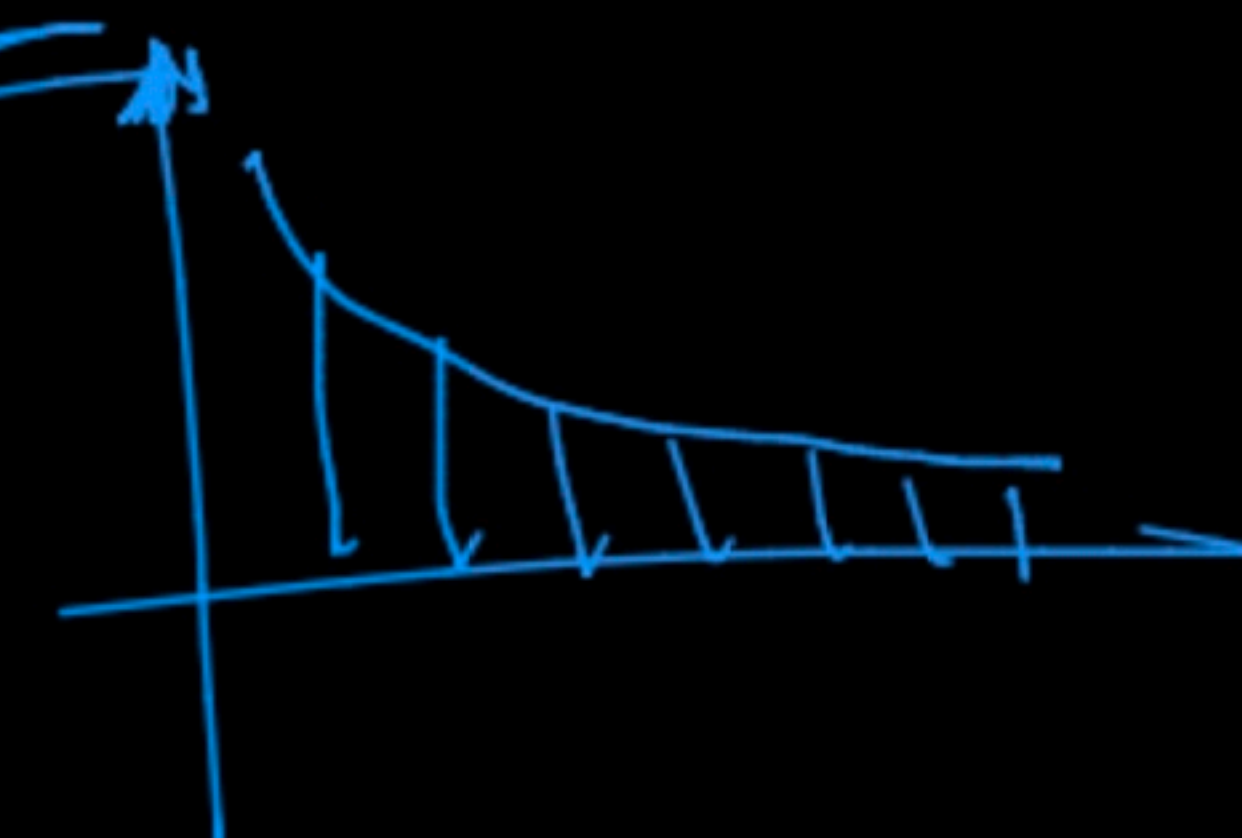
	x	y	$\hat{y}$	$(y - \hat{y})^2$
8	158	154	150	16
11	165	155	153	4
25	175	154	156	4
15	183	140	130	100

$\frac{SS}{4}$
MSE

$\overline{MAE}$

14
4

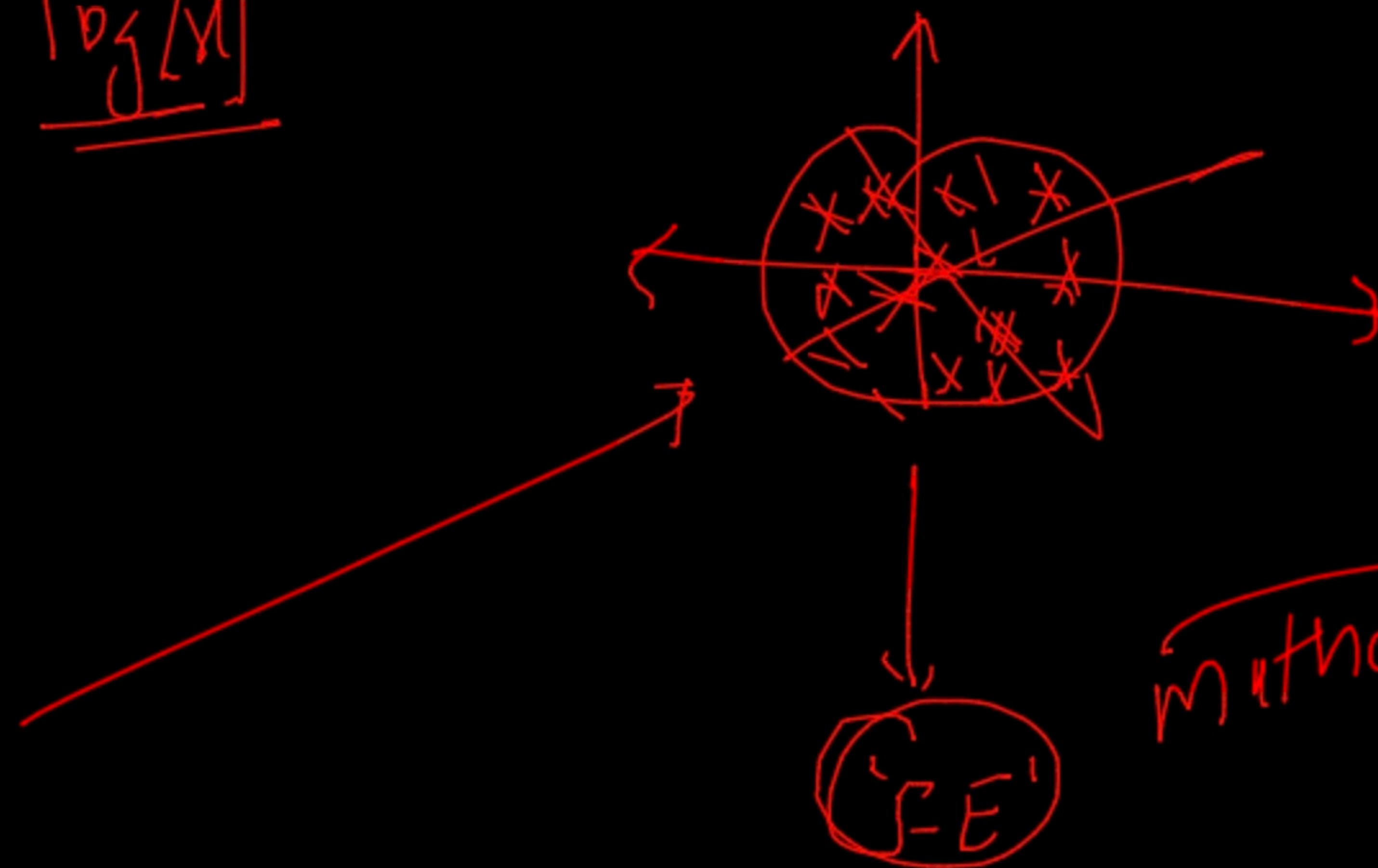
MSE



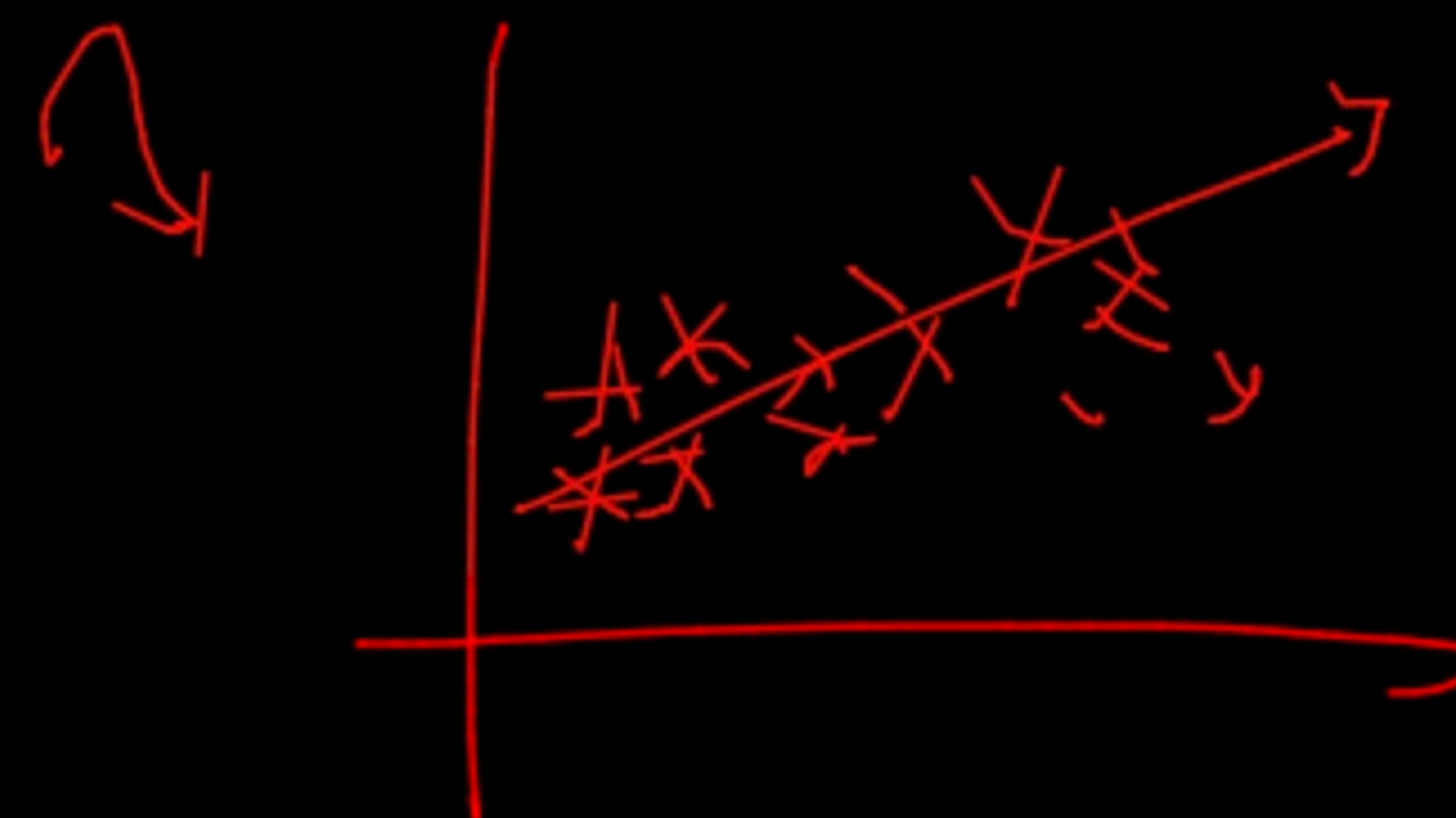
X Y $\log[M]$

'SUM'

kernel



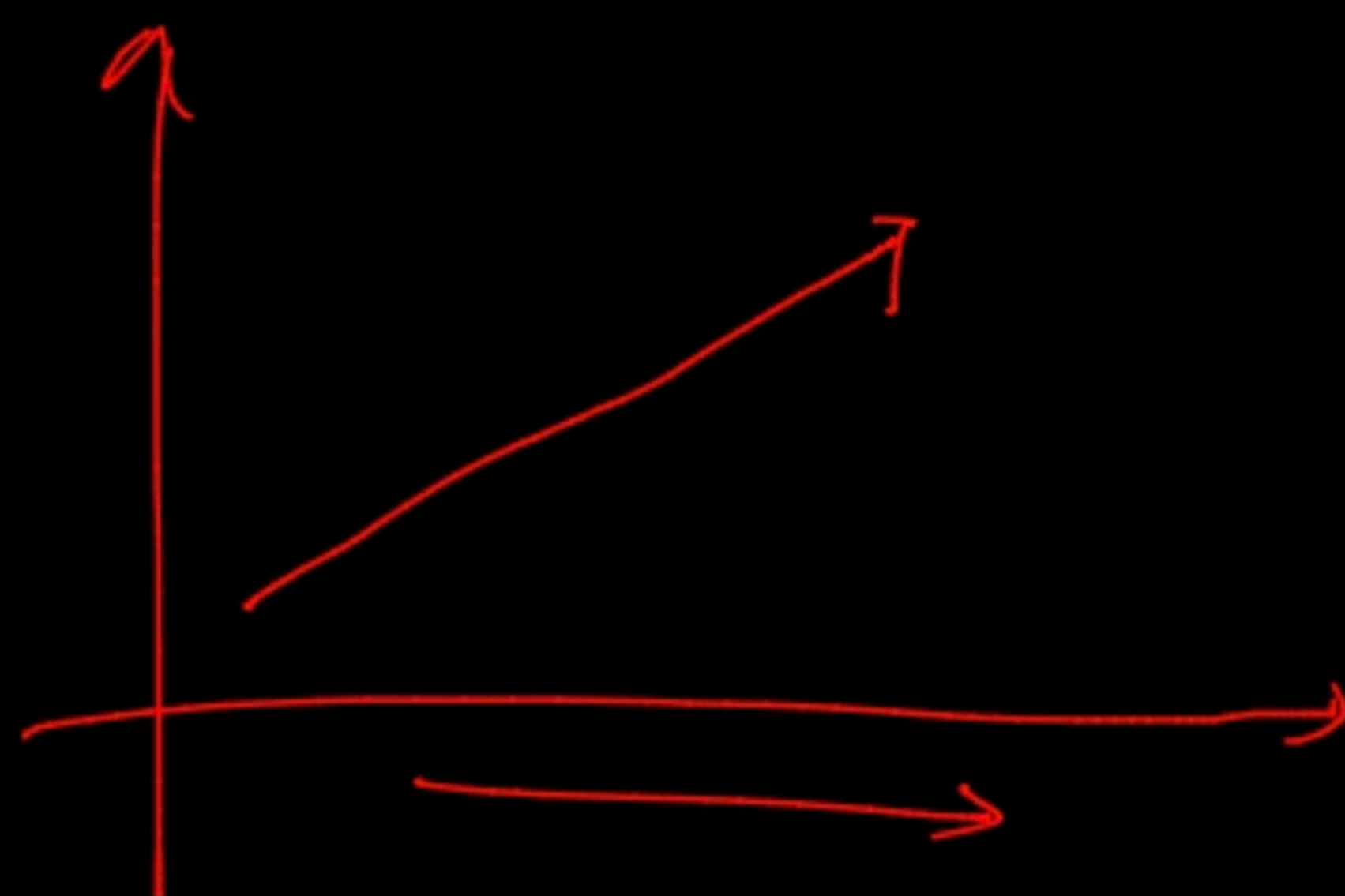
mathematical transformation



Salary

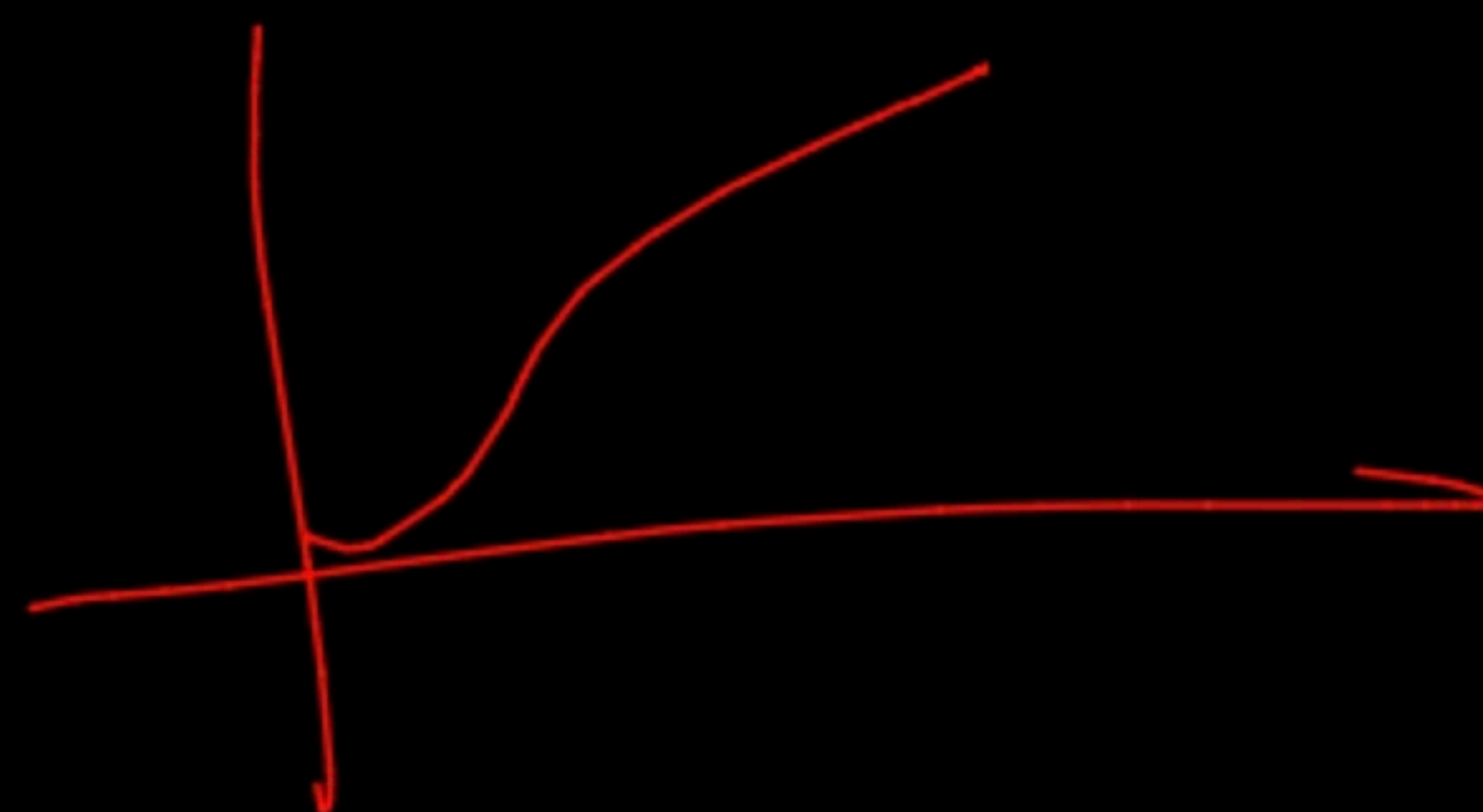
Ent

Salary



↓

log



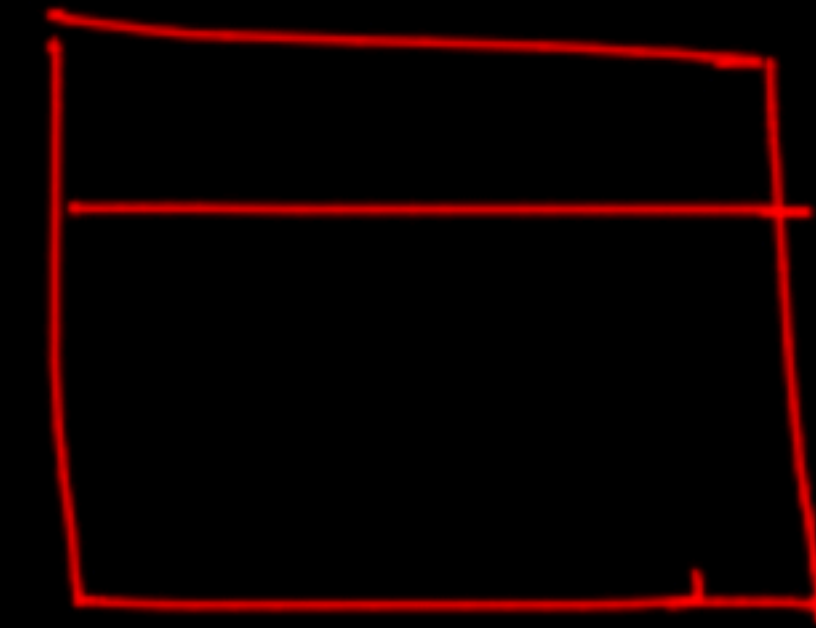
2D $\rightarrow$ line

$\downarrow$

3D $\rightarrow$ plane

$\vdots$

nD $\rightarrow$ Hyperplane



$$y_i = w_1 x_i + w_2 x_2 + w_0$$

$\vdots$

prediction

$$y_i = \sum_{j=1}^n w_j^T \cdot x_i + w_0$$

minimize

Regularization

7

↑
overfitting and underfitting

↑ (1) & (2) Black

Blackboard